Assignment – 16

Creating Tables and Indexes.

1) Write a command that will enable a user to pull orders grouped by date out of the Orders table quickly.

```
mysql> create index index_orders on orders(odate);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> show indexes from orders;
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
| Table  | Non_unique | Key_name     | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_commen
t | Visible | Expression |
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
| orders |          1 | index_orders |            1 | Odate       | A         |           3 |     NULL |   NULL | YES  | BTREE      |         |
  | YES     | NULL       |
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
1 row in set (0.07 sec)
```

2) If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
mysql> show indexes from orders;
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
| Table  | Non_unique | Key_name     | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_commen
t | Visible | Expression |
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
| orders |          0 | PRIMARY      |            1 | Onum        | A         |           8 |     NULL |   NULL |      | BTREE      |         |
  | YES     | NULL       |
| orders |          1 | index_orders |            1 | Odate       | A         |           3 |     NULL |   NULL | YES  | BTREE      |         |
  | YES     | NULL       |
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
--+---------+------------+
```

3)Create an index that would permit each salesperson to retrieve his or her orders grouped by date quickly.

```
mysql>  create index index_odate on orders(snum, odate);
```

```
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------
-+---------+------------+
| Table  | Non_unique | Key_name     | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment
| Visible | Expression |
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------
-+---------+------------+
| orders |          0 | PRIMARY      |            1 | Onum        | A         |           8 |     NULL |   NULL |      | BTREE      |         |
| YES     | NULL       |
| orders |          1 | index_odate  |            1 | Snum        | A         |           5 |     NULL |   NULL | YES  | BTREE      |         |
| YES     | NULL       |
| orders |          1 | index_odate  |            2 | Odate       | A         |           7 |     NULL |   NULL | YES  | BTREE      |         |
| YES     | NULL       |
+--------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+--------------
```